



From PhET to Keyboarding

How We Built an LTI Tool That Turns Any Web Interactive into a
Grable Canvas Assignment



VIRTUAL ARKANSAS

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About Me

Data Science Specialist, Virtual Arkansas

Teacher for 10 years: Physics, Chemistry, Biology, Physical Science, Business, Computer Science

Coding for about 25 years

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VIRTUAL ARKANSAS



Are You Using AI to Build Interactives?

Show of hands

Who is using AI to build their own simulations, games or practice tools?

Now grade it

Keep your hand up if it puts a grade in your gradebook.



No grade in the gradebook

Someone copies scores by hand, or it does not count

No record of the work

The teacher cannot see what the student did

Another login

Or another vendor, with another contract

What Zest Solves

AI can build the interactive. On its own, it is just a web page.



What is Zest?

An LTI 1.3 tool we built at Virtual Arkansas. Any HTML/JavaScript interactive becomes a Canvas assignment, auto-graded or teacher-graded, or a live element on a page.

In Canvas

The teacher clicks Embed Interactive Content and picks or uploads a zest

Zest

Runs the activity from your own server and saves work as the student goes

Back in Canvas

The grade lands in the gradebook and the teacher gets a custom view in SpeedGrader



How It Started

We built Zest for our own courses first, one need at a time.

Along the way we tested it with PhET-style science simulations.

Step 1

Coding tools to replace an expensive system

Step 2

Our keyboarding class

Step 3

More on the way



1. Teacher

Embed it from the library and set the passage

2. Student

Types and submits. The grade shows up in the gradebook.

3. Teacher

Opens SpeedGrader and sees the keyboarding review, not a file

4. Student, another device

The work is still there

Keyboarding Practice

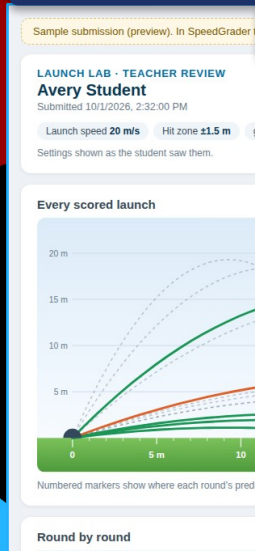
[Live demo](#)



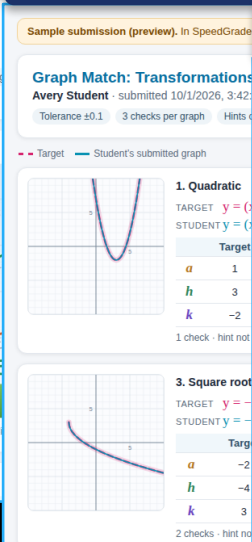
A Custom View in SpeedGrader

Each zest brings its own page for the teacher. SpeedGrader shows the actual work, not a file or a link.

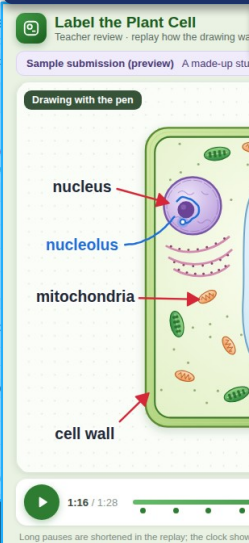
Launch Lab: every shot, on one field



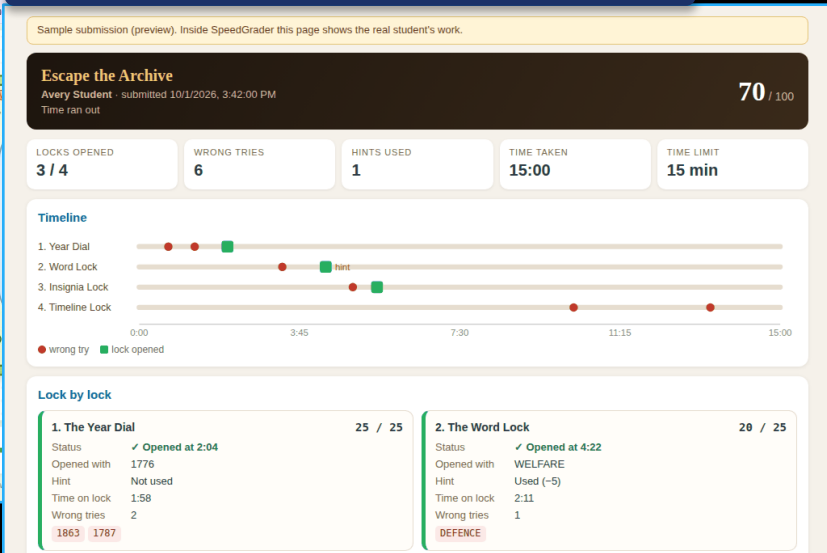
Graph Match: their curve over the target



Sketch & Label: replay of the drawing



Escape the Archive: every wrong try and hint



One Package, Lots of Courses

Set it up per course

A different passage for each class, from the same zest

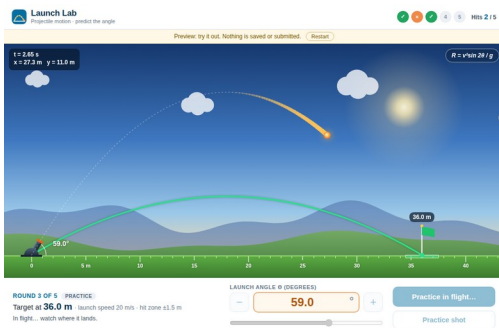
Fix it once, update every copy

No broken links and no lost student work

Works on managed Chromebooks

Even when third-party cookies are blocked

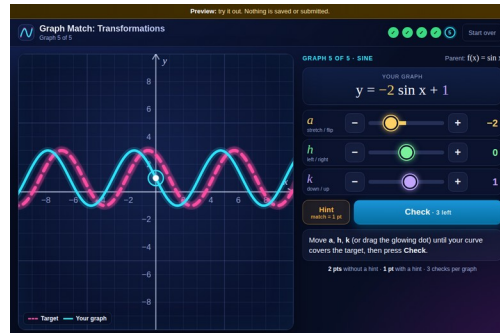




Launch Lab

Physics

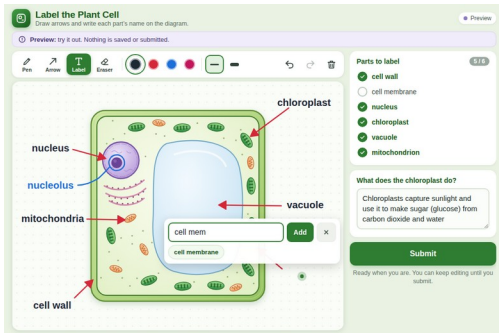
Auto-graded



Graph Match

Algebra 2

Auto-graded



Sketch & Label

Life Science

Teacher-graded

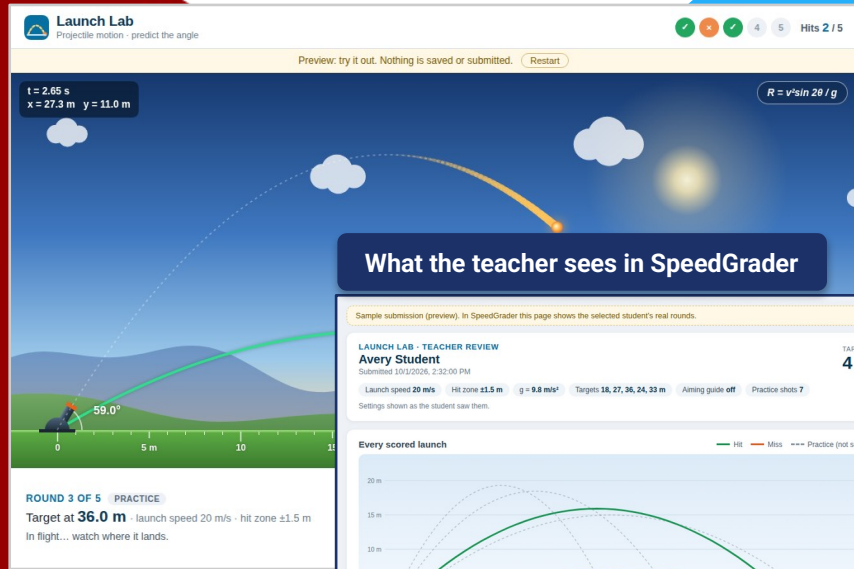


Escape the Archive

U.S. History

Auto-graded





What the teacher sees in SpeedGrader



Launch Lab

Physics | Auto-graded

The prompt

"Make a projectile motion lab where students predict the launch angle to hit a target, auto-graded, with the target distance set by the teacher."

Built from this sentence plus a few answers about rounds and practice shots.

QR code
(link
coming)



Graph Match

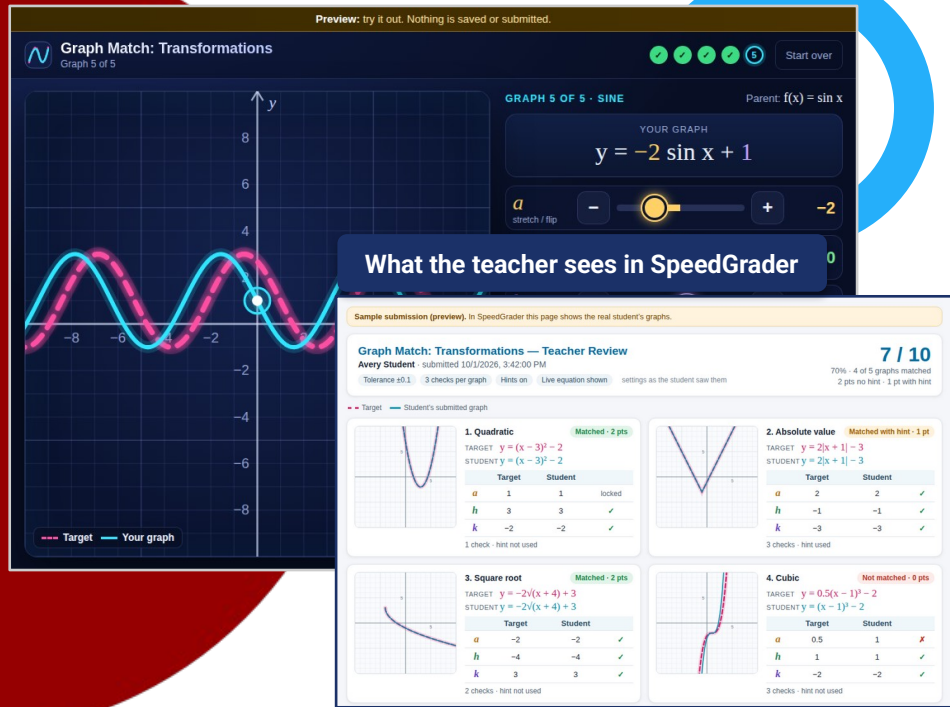
Algebra 2 | Auto-graded

The prompt

"Build a function transformations game for Algebra 2: students drag sliders to make their graph match a target curve. Auto-graded, and I want to pick the target functions myself."

Teachers pick a ready-made set of graphs, or build their own in a settings page the AI also built.

QR code
(link
coming)



Sketch & Label

Life Science | Teacher-graded

The prompt

"Students label a plant cell by drawing arrows and writing the names of the parts right on the diagram, then explain in a sentence what the chloroplast does. I'll grade it myself, and in SpeedGrader I want to see their drawing and watch how they drew it."

Students can draw with a finger on a phone or tablet.

QR code
(link
coming)

The image displays two overlapping screenshots of a digital learning interface. The top screenshot shows the 'Label the Plant Cell' activity with a drawing tool and a list of parts to label. The bottom screenshot shows the 'SpeedGrader' preview of a student's submission.

Label the Plant Cell
Draw arrows and write each part's name on the diagram.
Preview: try it out. Nothing is saved or submitted.

Parts to label
5/8
✓ cell wall
○ cell membrane
✓ nucleus
✓ chloroplast
✓ vacuole

What the teacher sees in SpeedGrader

Label the Plant Cell
Teacher review - replay how the drawing was made
Sample submission (preview) A made-up student, so you can try the replay. In SpeedGrader this shows the real student's work. ... Back to the activity

Drawing with the pen

AT A GLANCE
2m 51s
Time spent
8
Labels
7
Arrows
2 pen strokes - 1 erased

EXPLANATION
What does the chloroplast do?
Chloroplasts use energy from sunlight to make food (glucose) for the plant from carbon dioxide and water. This is called photosynthesis, and it gives off oxygen.
26 words

PARTS LABELLED
8/8
✓ cell wall
labelled "cell wall"
✓ cell membrane
labelled "cell membrane"
✓ nucleus
labelled "nucleus"
✓ chloroplast
labelled "chloroplast"
✓ vacuole
labelled "vacuole"

Long pauses are shortened in the replay; the clock shows the student's real time. Dots mark labels, red marks show erasing.



Escape the Archive

U.S. History | Auto-graded

The prompt

"Create a digital escape room for U.S. History where students unlock four locks using clues from primary sources, one of them about the Little Rock Nine. Auto-graded, with hints I can turn off."

Hints cost points. The teacher can turn them off or set a time limit.

QR code
(link
coming)

Preview: try it out. Nothing is saved or submitted.

Escape the Archive

U.S. HISTORY · PRIMARY SOURCES

OPEN 2/4 TIME 0:05

"Mob rule cannot be allowed to override the decisions of our courts."

DOCUMENT SUMMARY (PARAPHRASE, NOT A QUOTATION)
In September 1957, nine Black students were to start classes at all-white Central High School in Little Rock, Arkansas, under a federal court's desegregation order. Governor Orval Faubus used the Arkansas National Guard to keep them out. On September 24, 1957, President Dwight D. Eisenhower signed Executive Order 10730, which placed the Arkansas National Guard under federal control and sent soldiers of the U.S. Army's 101st Airborne Division to Little Rock. On September 25, those soldiers

What the teacher sees in SpeedGrader

Sample submission (preview). Inside SpeedGrader this page shows the real student's work.

Escape the Archive

Avary Student submitted 10/1/2026, 3:42:00 PM
Time ran out

70 / 100

LOCKS OPENED	WRONG TRIES	HINTS USED	TIME TAKEN	TIME LIMIT
3 / 4	6	1	15:00	15 min

Timeline

1. Year Dial
2. Word Lock
3. Insignia Lock
4. Timeline Lock

0:00 3:45 7:30 11:15 15:00

● wrong try ■ lock opened

Lock by lock

1. The Year Dial 25 / 25	2. The Word Lock 20 / 25
Status ✓ Opened at 2:04	Status ✓ Opened at 4:22
Opened with 1776	Opened with WELFARE
Hint Not used	Hint Used (-5)
Time on lock 1:58	Time on lock 2:11
Wrong tries 2	Wrong tries 1
1863 1787	DEFENCE



One Sentence

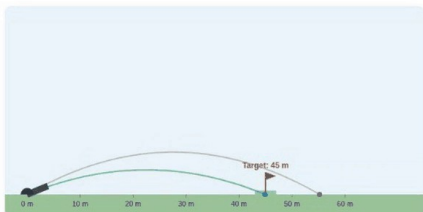
"Make a projectile motion lab where students predict the launch angle to hit a target, auto-graded, with the target distance set by the teacher."

That is all the teacher typed. About three minutes later, this.

Projectile Motion Lab: Hit the Target

A cannon on level ground fires a projectile at a fixed speed. Use what you know about projectile motion to **calculate** the launch angle that lands it on the target, then fire to test your prediction. Ignore air resistance.

● Unsaved changes



Your prediction

Launch angle (above horizontal)

22.4 °

Finished

Hint: on level ground the range is $R = v^2 \sin(2\theta) / g$. Solve it for θ . Two different angles give the same range, and either one counts.

You have 3 launches. Hit on launch 1 = 100 points, launch 2 = 80, launch 3 = 60, and so on (minimum 20). No hit = 0.

Score: 80 / 100

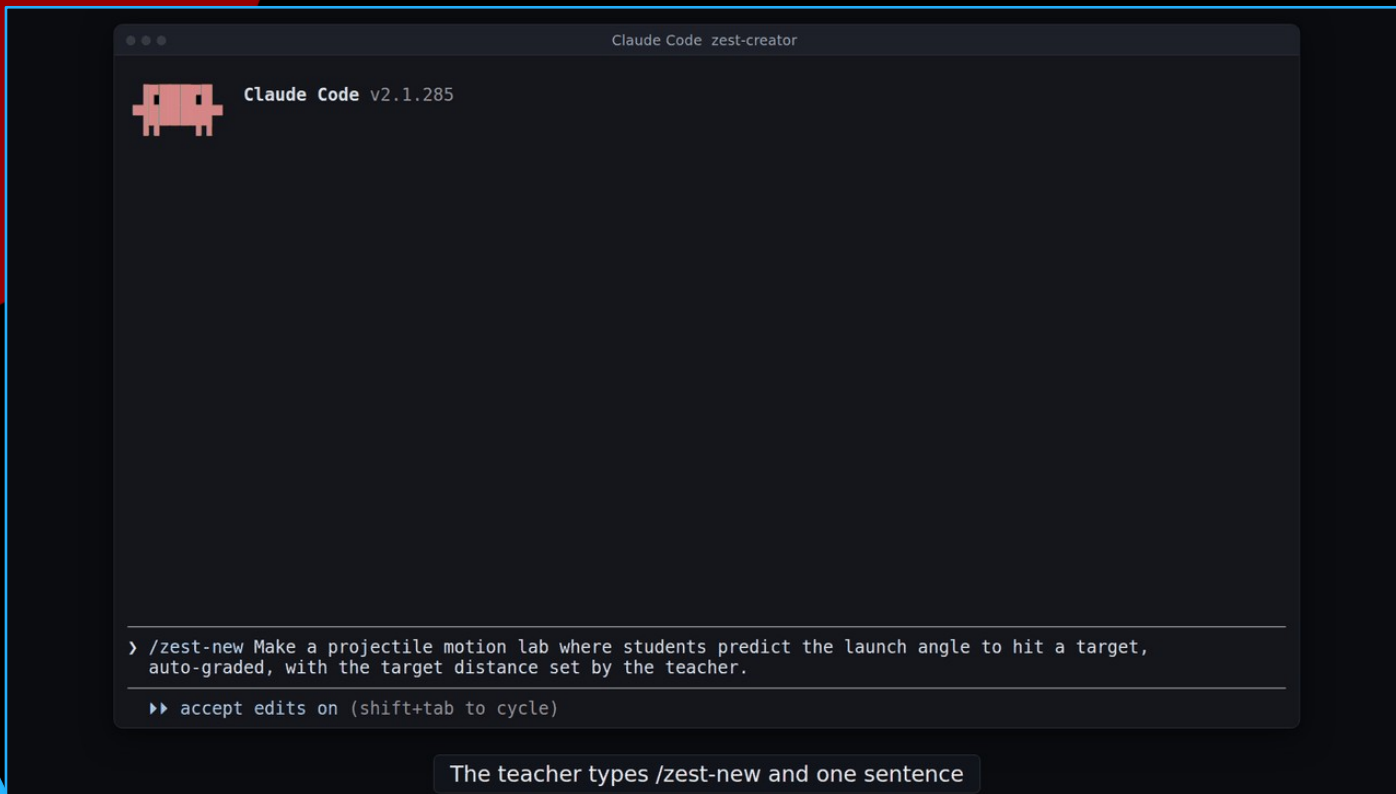
You hit the target on launch 2.

Submit score



Let's Watch

Real run, sped up



One File

A zest is a single file

Any program running Zest can upload it

Your keyboarding activity can be my keyboarding activity

Written Down

The package format is a published specification

github.com/VirtualArkansasOrg/zest-spec

The zest-creator template is public

github.com/VirtualArkansasOrg/zest-creator

Picture a shared library of gradable activities across programs



Challenges

What took the most work

LTI and Canvas

Most of the work. Again.

Canvas quirks

Pop-up alerts are blocked in assignments, and the frames can't resize

Chromebooks

Launches had to work with third-party cookies blocked

Never lose student work

Every change is tested with fake teachers, students and admins



Can You Trust What the AI Made?

What checks it, and what still needs a person

Checked

Every package is checked when it's built. A separate review looks for anything that sends data out or tracks students.

Sandboxed

Zest runs every package in a sandbox, with permissions off by default.

Still a person's job

Test it as a student. Have a subject expert check it. Use teacher grading for high stakes.



What's Next

Where Zest goes from here

1

Release the server code

MIT license

2

A separate domain for packages

The next security step

3

Accessibility review

No WCAG claims yet

4

Other LMSs

Researched Brightspace, Schoology and Buzz





forms.gle/1BDJ6hXbWxbZRopP8

Want the Code?

Scan to get the code when we release it
forms.gle/1BDJ6hXbWxbZRopP8

If you build zests, I want to trade
kyle.yancey@virtualarkansas.org

Thank you for your time 😊

"Zest" and "Zestable" are trademarks of Virtual Arkansas
(applications pending).





Get the code: forms.gle/1BDJ6hXbWxbZRopP8

Questions?

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